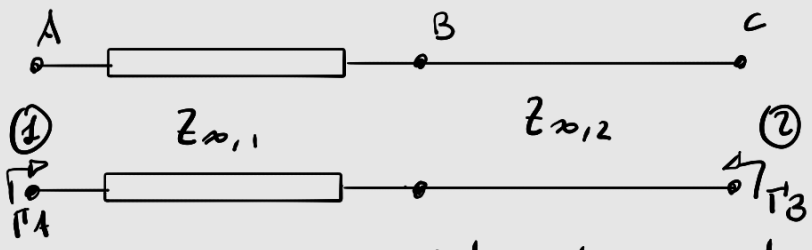
Ex. 1



$Z_{in} = Z_0$ for all the ports

$Z_{\infty_1} = 2Z_0$ $\qquad AB = \frac{7\lambda}{8}$

$Z_{\infty_2} = Z_0/3$ $\qquad BC = \frac{3\lambda}{4}$

$\Gamma_A = S_{11} \rightarrow$ we evaluate when impedence is matched at port 2

$\theta = \beta l$

- for section AB: $\theta_1 = \frac{2\pi}{\lambda} \cdot \frac{7\lambda}{8_4} = \frac{7}{4}\pi$ rad
- for section BC: $\theta_2 = \frac{2\pi}{\lambda} \cdot \frac{3\lambda}{4_2} = \frac{3}{2}\pi$ rad

Z_L on port 2 is equal to Z_0, we evaluate Γ_A

The BC line is a quarter wave transformer so

$$Z'_{in_B} = \frac{Z_{\infty_2}^2}{Z_0} = \frac{Z_0^2/9}{Z_0} = \frac{Z_0}{9}$$

$$Z_A = Z_{\infty_1} \frac{Z'_{in_B} + jZ_{\infty_1}\tan(\theta_1)}{Z_{\infty_1} + jZ'_{in_B}\tan(\theta_1)} = 2Z_0 \frac{\frac{Z_0}{9} + j2Z_0(-1)}{2Z_0 + j\frac{Z_0}{9}(-1)} = 2Z_0 \cdot \frac{1/9 - j2}{2 - j/9}$$

$$\Gamma_A = S_{11} = \frac{Z_A - Z_0}{Z_A + Z_0} = \frac{2Z_0 \frac{1/9 - j2}{2 - j/9} - Z_0}{2Z_0 \frac{1/9 - j2}{2 - j/9} + Z_0} = 0{,}55 - 0{,}73j$$

Z_L on port 1 is Z_0, we evaluate Γ_B

$$Z''_{in_B} = Z_{\infty_1} \frac{Z_0 + jZ_{\infty_1}\tan(\theta_1)}{Z_{\infty_1} + jZ_0\tan(\theta_1)} = 2Z_0 \frac{Z_0 + j2Z_0(-1)}{2Z_0 + jZ_0(-1)} = 2Z_0 \frac{1-2j}{2-j}$$

$$Z_C = \frac{Z_{\infty_2}^2}{Z''_{in_B}} = \frac{\frac{Z_0^2}{9}}{2Z_0\frac{1-2j}{2-j}} = \frac{Z_0}{9}\frac{1}{2\frac{1-2j}{2-j}} = \frac{Z_0}{18}\frac{2-j}{1-2j}$$

$$\Gamma_B = S_{22} = \frac{Z_C - Z_0}{Z_C + Z_0} = \frac{\frac{Z_0}{18}\frac{2-j}{1-2j} - Z_0}{\frac{Z_0}{18}\frac{2-j}{1-2j} + Z_0} = -0{,}91 + 0{,}061j$$

We use the definition of S_{21}:

$S_{21} = \frac{V_C}{V_A}(1 + S_{11})$ => when impedance matched on port ②, input node A and impedance at node B is =>

$\Rightarrow Z_{in\,B} = \frac{Z_0}{9}$

$I_B = \frac{V_B}{Z_0/9} = \frac{9V_B}{Z_0}$

$V_A = V_B \cos(\theta_1) + j\frac{9V_B}{Z_0} 2Z_0 \sin(\theta_1) = V_B \frac{\sqrt{2}}{2}(1 - j18)$

$\frac{V_B}{V_A} = \frac{\sqrt{2}}{1 - j18}$

We do the same for line BC

$I_C = \frac{V_C}{Z_0}$

$V_B = V_C \cos(\theta_2) + j\frac{V_C}{Z_0}\frac{Z_0}{3}\sin(\theta_2) = -j\frac{V_C}{3}$

$\frac{V_C}{V_A} = \frac{V_C}{V_B} \cdot \frac{V_B}{V_A} = -0{,}23 + 0{,}013j$

$S_{21} = \frac{V_C}{V_A}(1 + S_{11}) = -0{,}35 + 0{,}19j$

Since the line is completely passive then it's reciprocal, therefore we can write the S matrix as:

$$S = \begin{pmatrix} 0{,}55 - 0{,}73j & -0{,}35 + 0{,}19j \\ -0{,}35 + 0{,}19j & -0{,}91 + 0{,}061j \end{pmatrix}$$

Calculation of ABCD parameters

. For line AB

$A = \cos(\theta_1) = \frac{\sqrt{2}}{2} \simeq 0{,}71$

$B = jZ_{\infty,1}\sin(\theta_1) = -j\cancel{2}Z_0\frac{\sqrt{2}}{\cancel{2}} = -j\sqrt{2}Z_0$

$C = jY_{\infty,1}\sin(\theta_1) = j\frac{-1}{2Z_0}\frac{\sqrt{2}}{2} = -j\frac{\sqrt{2}}{4Z_0}$

$D = \cos(\theta_1) \simeq 0{,}71$

• For line BC

$A = \cos(\theta_2) = 0$

$B = j Z_{\infty,2} \sin(\theta_1) = -j \frac{Z_0}{3}$

$C = j Y_{\infty,1} \sin(\theta_1) = -j \frac{3}{Z_0}$

$D = \cos(\theta_2) = 0$

$$M_{TOT} = \begin{pmatrix} 0,71 & -j\sqrt{2}\, Z_0 \\ -j\frac{\sqrt{2}}{4Z_0} & 0,71 \end{pmatrix} \begin{pmatrix} 0 & -j\frac{Z_0}{3} \\ -j\frac{3}{Z_0} & 0 \end{pmatrix} = \begin{pmatrix} -4,24 & -j\,0,24\,Z_0 \\ -j\frac{2,13}{Z_0} & -0,12 \end{pmatrix}$$

The Den. is common to all parameters, so:

$$Den = Z_0 A_{TOT} + B_{TOT} + Z_0^2 C_{TOT} + Z_0 D_{TOT} = -4,36 - 2,36j$$

$$S_{11} = \frac{Z_0 A_{TOT} + B_{TOT} - Z_0^2 C_{TOT} - Z_0 D_{TOT}}{Den} = 0,55 - 0,73j$$

$$S_{12} = \frac{2Z_0 (A_{TOT} D_{TOT} - B_{TOT} C_{TOT})}{Den} = \frac{2Z_0}{Den} = -0,35 + 0,19j$$

$$S_{21} = \frac{2Z_0}{Den} = -0,35 + 0,19j$$

$$S_{22} = \frac{-Z_0 A_{TOT} + B_{TOT} - Z_0^2 C_{TOT} + Z_0 D_{TOT}}{Den} = -0,91 + 0,061j$$

The S matrix evaluated with the ABCD parameters is:

$$S = \begin{pmatrix} 0,55 - 0,73j & -0,35 + 0,19j \\ -0,35 + 0,19j & -0,91 + 0,061j \end{pmatrix}$$

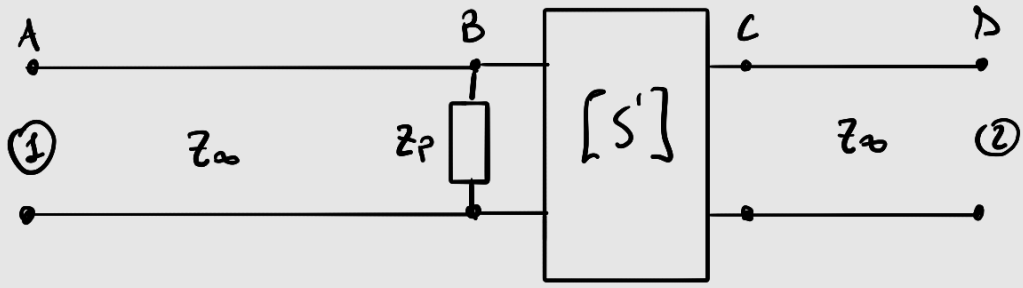
Questions:

① The entire network is reciprocal because it is passive and because $S_{12} = S_{21}$

② It isn't symmetrical because $S_{11} \neq S_{22}$

③ The port is lossless because $|S_{11}|^2 + |S_{21}|^2 = 0{,}994 \approx 1$.

Ex 2.



$Z_{in} = Z_0$ → even for the device between B and C

② Closed on reference impedence ⇒ matched!

$Z_0 = 50\Omega$ $\quad Z_\infty = 150\Omega$ $\quad Z_P = 50\Omega$

$AB = 2\lambda$ $\quad CD = \frac{3\lambda}{2}$

$$S' = \begin{bmatrix} 0{,}8 & 0{,}5j \\ 0{,}2j & 3 \end{bmatrix}$$

Notes: AB and CD are both multiples of half wavelenght so we can say that

$V_B = V_A$ → full phase shift at every $\lambda/2$

$V_D = -V_C$ → half phase shift, so it's negative.

Port ② matched ⇒ Z_0 at port ②

$S'_{11} = 0{,}8$ → Γ'_B from the port looking into blok S'

$$Z'_{in} = Z_0 \frac{1+S'_{11}}{1-S'_{11}} = 50 \frac{1+0{,}8}{1-0{,}8} = 450\Omega$$ → CD line "transparent" to impedence

$$Z_{B_{TOT}} = Z_P // Z'_{in} = \left(\frac{1}{Z_P} + \frac{1}{Z'_{in}}\right)^{-1} = 45\Omega$$

Since AB line is transparent to impedence $Z_A = Z_{B_{TOT}}$

$$S_{11} = \frac{Z_A - Z_0}{Z_A + Z_0} = \frac{45-50}{45+50} = -0{,}053$$

Now we track the voltage from port ① to ②

The voltage at node A and B is equal!! (2λ TL lenght)

$V_A = V_B = a_1\sqrt{Z_0}\,(1+S_{11}) = 6{,}69 \cdot a_1$

Now for V_B in respect of the waves inside the block, which have to be equal

$V_B' = a_1\sqrt{Z_0}\,(1+S_{11}') = a_1'\sqrt{50}\,(1+0{,}8) = 12{,}73 \cdot a_1'$

↳ seen from node B

$V_B = V_{B'} \Rightarrow 6{,}69\, a_1 = 12{,}73 \cdot a_1' \Rightarrow a_1' = 0{,}53\, a_1$

Node D is terminated with Z_0 ⇒ CD transport to impedance ⇒ Node C is terminated with Z_0, so there is no wave reflecting into the block.

$b_2' = S_{21}'\, a_1' = 0{,}2j \cdot 0{,}53\, a_1 = 0{,}11j\, a_1$

$b_2 = -b_2' = -0{,}11j\, a_1$

↳ CD's lenght is $\frac{3\lambda}{2}$ so the voltage "is flipped"

$$\left.\frac{b_2}{a_1}\right|_{\text{port 2 matched}} = \frac{-0{,}11j a_1}{a_1} = -0{,}11j = S_{21}$$

↳ We apply the definition

To compute the P_2/P_1 ratio, since $\Gamma_D' = 0$ because of impedance matching, then:

$$\frac{P_2}{P_1} = \frac{|S_{21}|^2}{1-|S_{11}|^2} = \frac{|-0{,}11j|^2}{1-|-0{,}053|^2} = 0{,}012$$